Basic background

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Goal

- This module provides an overview of the basic background used in the Software Security Area. The module is structured in two lectures.
- In this first lecture we will learn…
  - how binary files are stored;
  - the basic steps that transform a source code into programs;
  - principles of x86 architecture;
  - mechanisms used to support memory management.

Prerequisites

- Lecture:
  - Basic knowledge of C
  - Basic knowledge of Python

Outline

- From Code to Programs
- Executable and Linkable Format (ELF)
- x86 architectures
- Memory management and Calling conventions
- Debugging

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From code to programs

- Compiling a C program is a multi-stage process composed of four steps:
  - preprocessing
  - compilation
  - assembly
  - linking

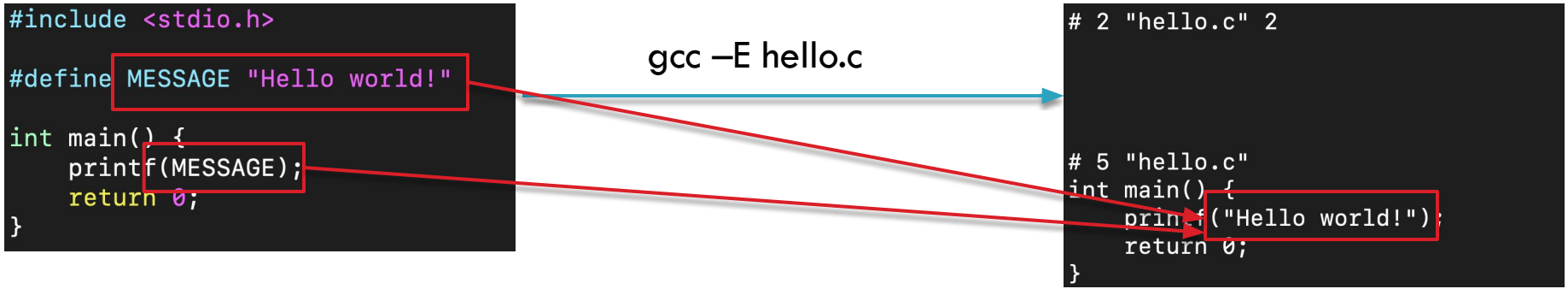
GCC Compiler

- In this lecture *gcc* compiler will be used…
  - is the C compiler included in the *GNU Compiler Collection*
  - is available in the main operating systems
    - Standard compiler in all the UNIX/Linux distributions
- Detailed documentation is available at the following link:

https://gcc.gnu.org/onlinedocs/

From code to programs: preprocessing

- In the first phase *preprocessor commands (in C they start with '#') are interpreted:*



From code to programs: compilation

- In the second phase, preprocessed code is translated into *assembly instructions*:

```c
#include <stdio.h>

#define MESSAGE "Hello world!"

int main() {
    printf(MESSAGE);
    return 0;
}
```

gcc –s hello.c

```
# 2 "hello.c" 2



# 5 "hello.c"
int main() {
    printf("Hello world!");
    return 0;
}
```

```
main:
.LFB0:
        .cfi_startproc
        pushq   %rbp
        .cfi_def_cfa_offset 16
        .cfi_offset 6, -16
        movq    %rsp, %rbp
        .cfi_def_cfa_register 6
        movl    $.LC0, %edi
        movl    $0, %eax
        call    printf
        movl    $0, %eax
        popq    %rbp
        .cfi_def_cfa 7, 8
        ret
```

From code to programs: assembly

- In the *assembly* phase assembly instructions are translated into *machine* or *object code*:

From code to programs: linking

- In the last phase (multiple) *object code* are combined in a single executable
- In the generated file, references (links) to the used library are added

Static vs Dynamic Linking

- Two approaches can be used in the linking phase:
  - Static Link
    - Binaries are *self-contained* and do not depend on any external libraries
  - Dynamic Link
    - Binaries rely on system libraries that are loaded when needed
    - Mechanisms are needed to *dynamically* relocate code

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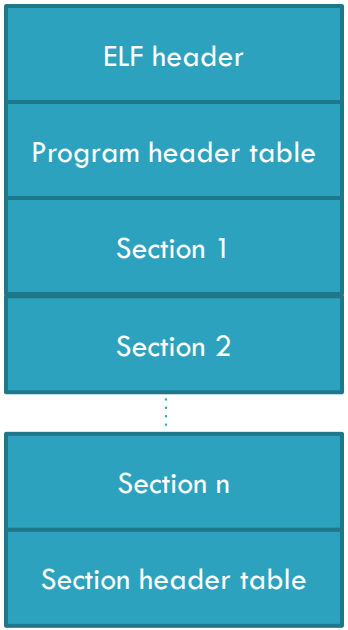
Executable and Linkable Format

Non entriamo nel dettaglio

- The Executable and Linkable Format (ELF) is a common file format for object files
- There are three types of object files:
  - Relocatable file containing code and data that can be linked with other object files to create an executable or shared object file
  - **Executable files** holding a program suitable for execution
  - Shared object files that can be:
    - linked with other relocatable and shared object files to obtain another object file
    - used by a dynamic linker together with other executable files and object files to create a process image

Executable and Linkable Format

- Any ELF file is structured as:
  - an ELF header describing the file content
  - a Program header table providing info about how to create a process image
  - a sequence of Sections containing what is needed for linking (instructions, data, symbol table, relocation information,…)
  - a Section header table with a description of previous sections



Non entriamo nel dettaglio

ELF: Relevant sections

- .text: contains the executable instructions of a program
- .bss: contains uninitialised data that contribute to the program's memory image
- .data, .data1: contain initialized data that contribute to the program's memory image
- .rodata, .rodata1: are similar to .data and .data1, but refer to read-only data
- .symtab: contains the program's symbol table
- .dynamic: provides linking information

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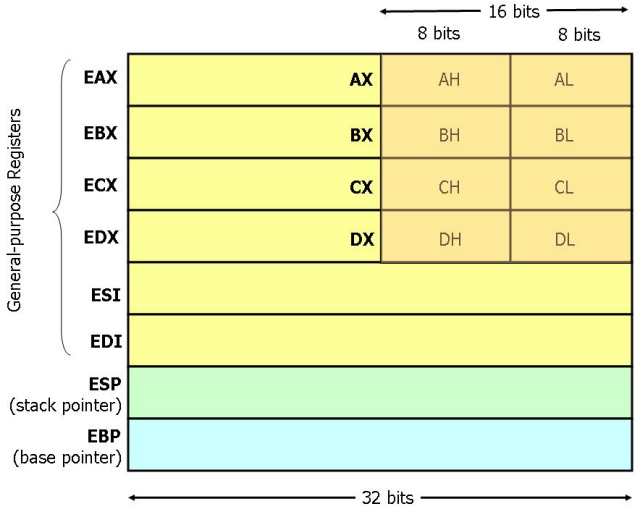
x86 Instruction Sets

- We provide a short introduction of a small but useful subset of the available instructions and assembler directives of x86 assembly language

- A detailed description can be found at the following links:
  - Intel x86 Instruction Set Reference
    - http://www.felixcloutier.com/x86/
  - Intel's Pentium Manuals
    - http://www.intel.com/content/www/us/en/processors/architectures-software-developer-manuals.html

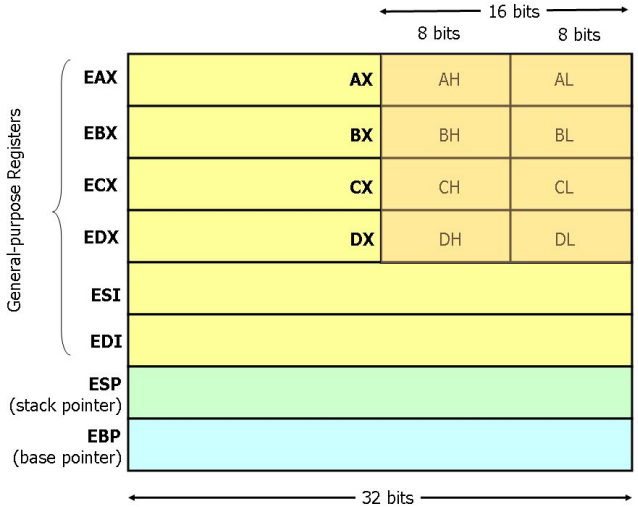
X86-32 Registers

- x86-32 processors have eight 32-bit general purpose registers
- The register names are mostly historical…
  - *EAX* used to be called *the accumulator* since it was used by a number of arithmetic operations
  - *ECX* was known as the *counter* since it was used to hold a loop index
- Two are reserved for special purposes:
  - the *stack pointer* (ESP)
  - the *base pointer* (EBP)



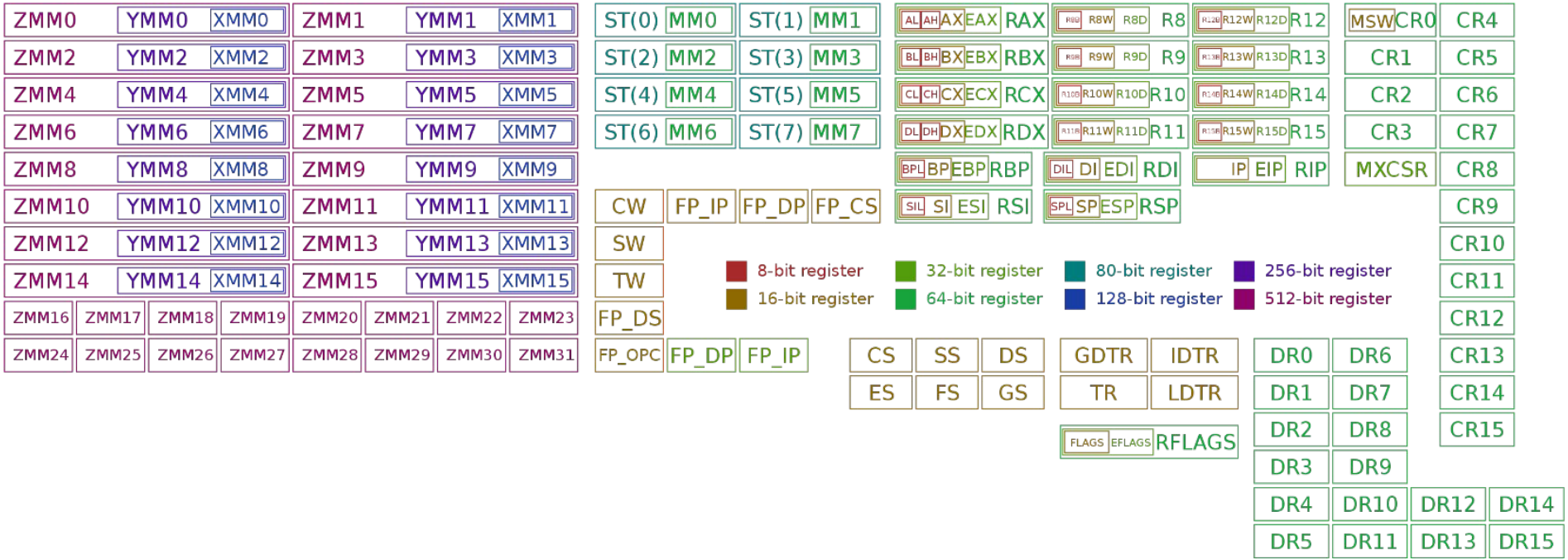
X86-32 Registers

- For the *EAX*, *EBX*, *ECX*, and *EDX* registers, subsections may be used
- For example:
  - *AX* refers to the least significant 2 bytes (16 bits) of EAX
  - *AL* refers to the least significant byte (8 bits) of *AX*
  - *AH* refers to the most significant byte (8 bits) of *AX*
- These sub-registers are mainly hold-overs from older, 16-bit versions of the instruction set



X86-64 Registers

- X86-64 architecture provides a larger set of registers



x86 Instructions

- Data Movement Instructions
  - *mov op1,op2:* copies the data item referred to by *op1* into the location referred to by *op2*
  - *push op1:* places its operand onto the top of the stack
  - *pop op1:* removes the 4-byte data element from the top of the stack
  - *lea op1,op2*: load the memory address indicated by *op2* into the register specified by *op1*

x86 Instructions

- Arithmetic and Logic Instructions
  - *add op1,op2:* stores in *op1* the result of *op2+op1*
  - *sub op1,op2:* stores in *op1* the result of *op2-op1*
  - *and op1,op2*
  - *or op1,op2*
  - *xor op1,op2*

  Perform the specified logical operation on the operands, storing the result in the first operand location
  - *…*

x86 Instructions

- Control Flow Instructions
  - *jmp op:* jump to the instruction at the memory location specified by the operand *op*
  - *cmp op1,op2:* compares the values of the two specified operands and stores the result in the *machine status word*
  - *j<condition> op:* depending on the *<condition>* and on the context of *machine status word*, jumps to instruction at the memory location indicated by the operand
    - *For example, jz (Jump if Zero), jc (Jump if Carry), and jo (Jump if Overflow) depend on specific flags.*

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Memory management

- Many of the modern programming languages allow programmers to use data types without having to know how they are represented
- Similarly, programmers often ignore where data is stored (allocated)…
  - compiler makes decisions, however at runtime allocation is under the control of Operating System and CPU

Memory management

- In some programming languages, like C, memory management can be controlled by programmers:
  - memory can be dynamically allocated and deallocated
  - memory address of variables can be obtained (pointers)
- If *x* is a variable, *&x* denotes the pointer to x, i.e., the memory address where x is stored

Memory allocation…

- Let us consider the following simple C program:

```c
#include <stdio.h>

int main() {
    int i;
    char c;
    short s;
    long l;

    printf("i is allocated at %p\n", &i);
    printf("c is allocated at %p\n", &c);
    printf("s is allocated at %p\n", &s);
    printf("l is allocated at %p\n", &l);
}
```

We can assume that:

- *int* needs 4 bytes
- *char* needs 1 byte
- *short* needs 2 bytes
- *long* needs 8 bytes

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    printf("l is allocated at %p\n", &l);
}
```

Variable declarations

We can assume that:

- *int* needs 4 bytes
- *char* needs 1 byte
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Memory allocation…

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    printf("s is allocated at %p\n", &s);
    printf("l is allocated at %p\n", &l);
}
```

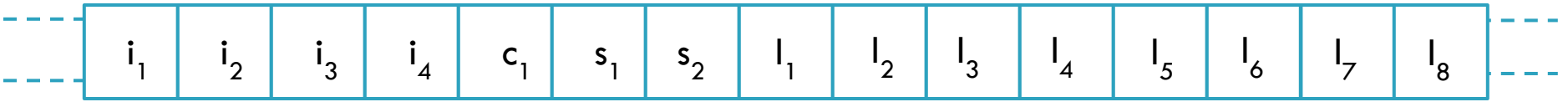
Variable addresses

We can assume that:

- *int* needs 4 bytes
- *char* needs 1 byte
- *short* needs 2 bytes
- *long* needs 8 bytes

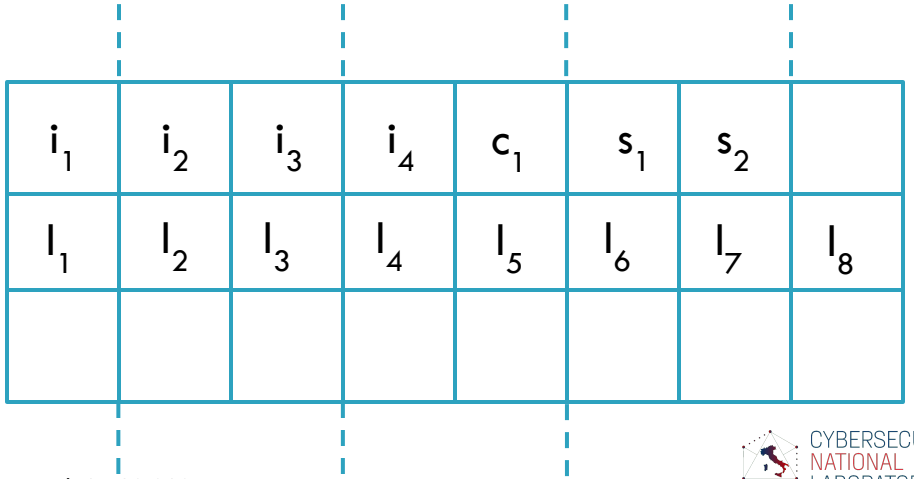
Memory allocation…

- Memory is usually represented as a sequence of bytes:



In a n·8 architecture, bytes are arranged in groups of n:

i ⟶ int
c ⟶ char
s ⟶ short
l ⟶ long



Memory allocation…


```c
#include <stdio.h>

int main() {
    int i;
    char c;
    short s;
    long l;

    printf("i is allocated at %p\n", &i);
    printf("c is allocated at %p\n", &c);
    printf("s is allocated at %p\n", &s);
    printf("l is allocated at %p\n", &l);
}
```

We can run this simple program to observe how data are allocated in memory

Memory allocation…

-O (oppure -O1): *"Optimize"*

-O2: *"Optimize even more"*

- Different allocation policies can be used by *gcc*:

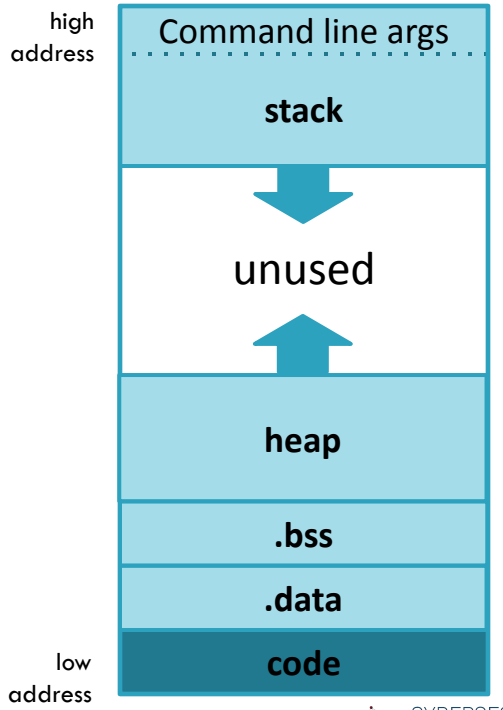
```
CC> gcc -o alignment alignment.c
CC> ./alignment
i is allocated at 0x7ffee117e7cc
c is allocated at 0x7ffee117e7cb
s is allocated at 0x7ffee117e7c8
l is allocated at 0x7ffee117e7c0
CC>
```

```
CC> gcc -o alignment alignment.c -O2
CC> ./alignment
i is allocated at 0x7ffee4dbb7c8
c is allocated at 0x7ffee4dbb7cf
s is allocated at 0x7ffee4dbb7cc
l is allocated at 0x7ffee4dbb7c0
CC>
```

- Compilers may introduce padding or change the order of data in memory
- There are trade-offs between speed and memory usage

Memory segments

- Memory is allocated for each process (a running program) to store data and code.
- This allocated memory consists of different segments:
  - *stack:* for local variables (grows *downward*)
  - *heap:* for dynamic memory (grows *upward*)
  - *data segment:*
    - *global uninitialized variables (.bss)*
    - *global initialized variables (.data)*
  - *code segment*

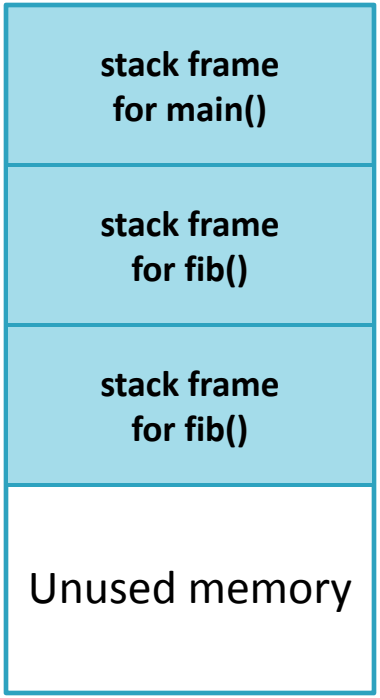


The stack

- The stack consists of a sequence of *stack frames* (or activation records), each for each function call:
  - allocated on *call*
  - de-allocated on *return*

```
int main(int argc, char **argv) {
  int f = fib( n: 10);
  printf("FIB(10)=%d\n",f);
}

int fib(int n) {
  int f1;
  int f2;
  if (n<=2) {
      return 1;
  } else {
      f1 = fib( n: n-1);
      f2 = fib( n: n-2);
      return f1+f2;
  }
}
```



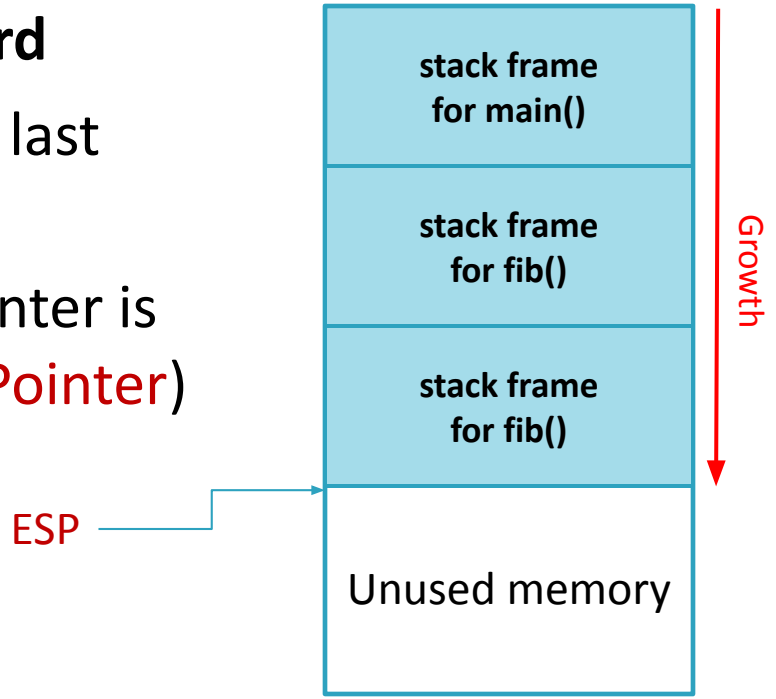
The stack

- The precise structure and organization of the stack depends on system architecture, operating system, and compilers we are using.
- For the sake of simplicity, in this lecture we will focus on *x86* architectures (32 and 64 bits) and on *gcc* compiler on *Linux*

- More detailed information are available at the following links:
  - https://docs.microsoft.com/en-us/cpp/cpp/argument-passing-and-naming-conventions
  - http://refspecs.linuxbase.org/

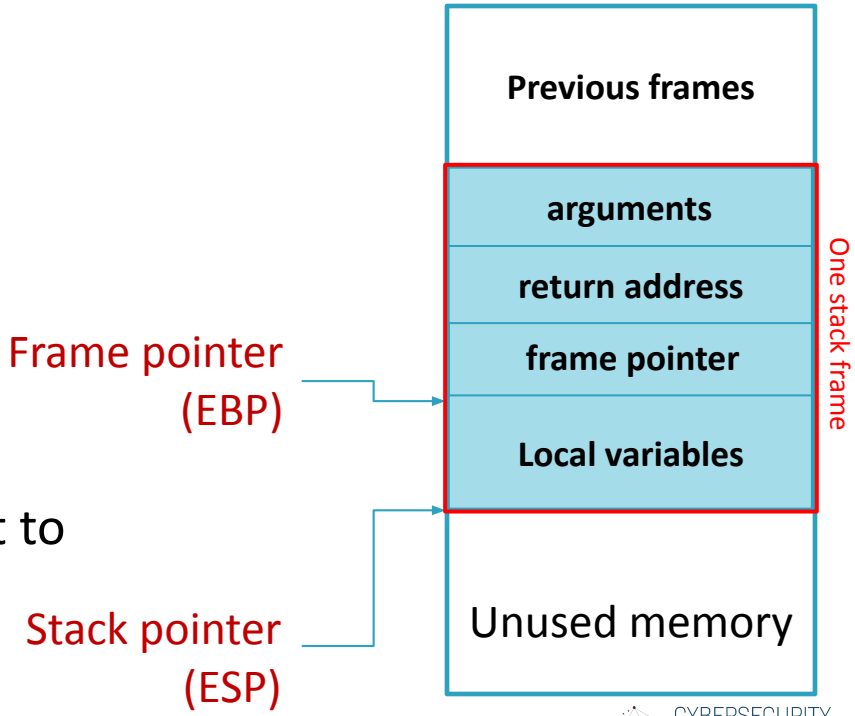
The stack

- Typically, the stack grows **downward**
- The stack pointer (SP) refers to the last element on the stack
- On *x86* architectures, the stack pointer is stored in the ESP (Extended Stack Pointer) register



Stack frame (for x86)

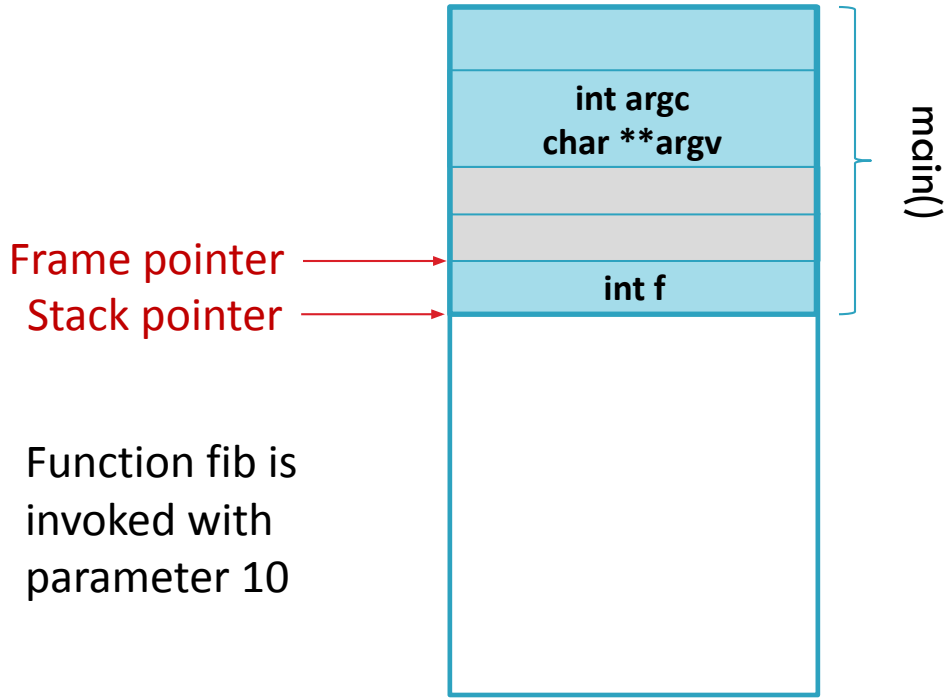
- In *x86* architecture, each stack frame contains:
  - Function arguments
  - Local variables
  - Copies of registries that must be restored:
    - return address
    - previous frame pointer
- Frame pointer, named Extended Base Pointer (EBP), provides a starting point to local variables



Stack frame: example


```
int main(int argc, char **argv) {
  int f = fib( n: 10);
  printf("FIB(10)=%d\n",f);
}

int fib(int n) {
  int f1;
  int f2;
  if (n<=2) {
      return 1;
  } else {
      f1 = fib( n: n-1);
      f2 = fib( n: n-2);
      return f1+f2;
  }
}
```



Function fib is invoked with parameter 10

Stack frame: example


```
int main(int argc, char **argv) {
  int f = fib( n: 10);
  printf("FIB(10)=%d\n",f);
}

int fib(int n) {
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  if (n<=2) {
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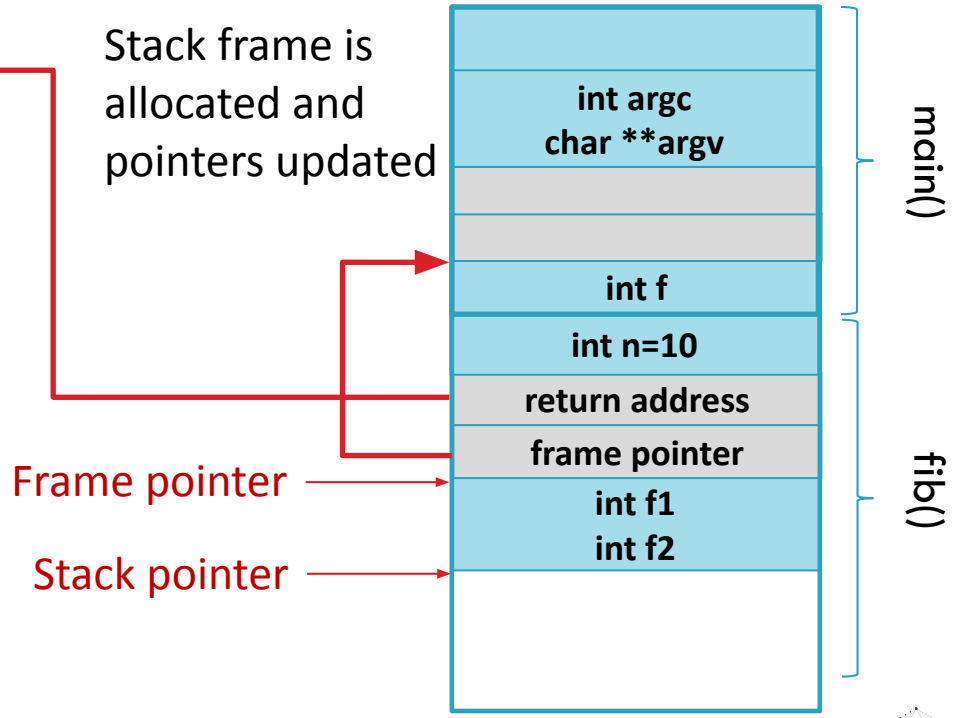

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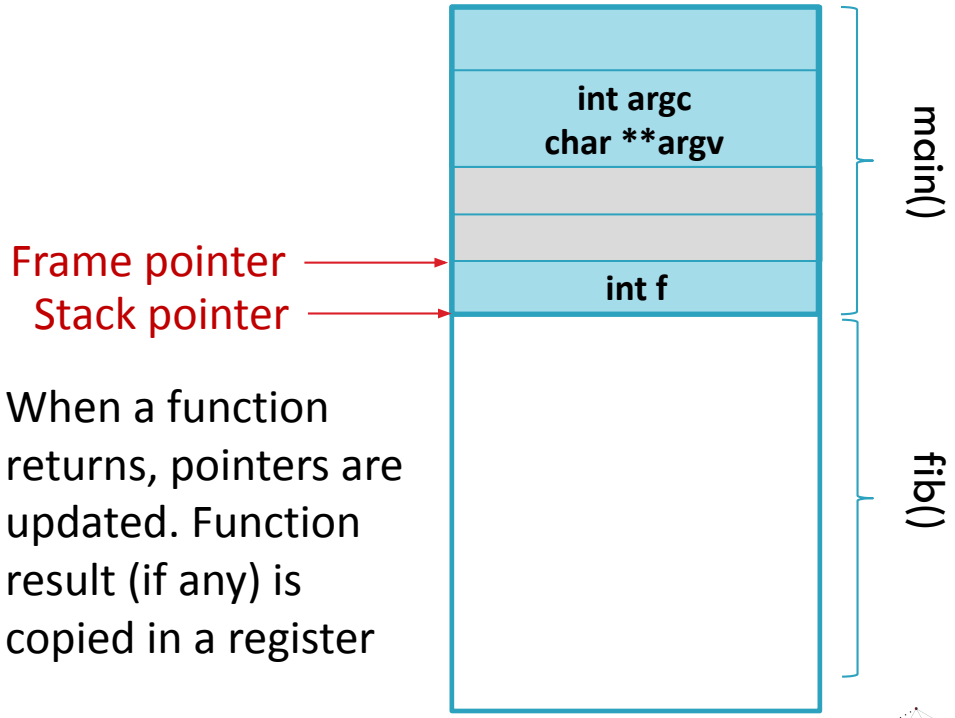

When a function returns, pointers are updated. Function result (if any) is copied in a register

Stack frame (for x86-64)

Non entriamo nel dettaglio

- In the *64-bit* version of *x86:*
  - Registers are extended to 64 bits
  - A new set of 8 registers is added
- The arguments are not all in the stack:
  - The first 6 are passed via registers
  - The remaining are placed in the stack (like for *x86*)
- Pointers *EBP* and *ESP* are named *RBP* and *RSP*
- A *red zone* of 128 bytes is placed in the stack just under RSP
  - This can be used to store extra local variables and is not modified by interrupt/exception/signal handlers

C declaration: cdecl

Non entriamo nel dettaglio

- The cdecl (C declaration) is a calling convention used in many compilers for *x86*
  - A calling convention describes how functions receive their parameters from caller and how they return they results
- In cdecl **arguments are passed on the stack**, while **values are returned via registers**
- Function arguments are pushed on the stack in the *right-to-left order*
- The caller *cleans* the stack after the RETURN from the called function

The heap

Non entriamo nel dettaglio

- Memory allocation and de-allocation in the stack is very fast
  - However, this memory cannot be used after a function returns
- The heap is used to store dynamically allocated data that outlive function calls:
  - This area is under programmer's responsibility

Memory management functions

Non entriamo nel dettaglio

- Basic *C* functions for memory management are:
  - *malloc(int),* given an integer *n* allocates an area of *n* (continuous) byes and returns a pointer to that area
  - *free(void*)*, deallocates the memory associated with a pointer

Stack and Heap pointers

Non entriamo nel dettaglio

- We have always to preserve reference to allocated areas in the heap
  - Otherwise, we cannot use them!
- These references (pointers) can be stored in the stack or in the heap
- We could store pointers to the stack in the heap
  - This can be dangerous! Referenced memory could be released!
  - Thus, the stack pointer is located at the top of the stack

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Debugging

Non entriamo nel dettaglio

- Debugging is the process of finding and fixing bugs (defects or problems that prevent correct operation) within computer programs, software, or systems.
- Debugging can be based on different techniques and methodologies such as:
  - interactive debugging
  - control flow analysis
  - unit and integration testing
  - log file and output analysis
  - memory dumps
  - profiling
- Many programming languages and software development tools also offer programs to aid in debugging, known as *debuggers*

- A debugger is a software tool that allows to:
  - run the target program under controlled conditions
  - track program operations in progress
  - monitor changes in computer resources
  - display the contents of memory
  - modify memory or register contents

Debugging

Non entriamo nel dettaglio

- Compilers can be instrumented to emit extra (optional) data that a debugger can use for a more informative input
- In the case of *gcc* compiler, the parameter *–g* can be used
- Debugging information is stored in specific sections of ELF file:
  - .debug, containing info for symbolic debugging
  - .line, containing line number information

Example

Non entriamo nel dettaglio


```
CC> gcc -o hello -g hello.c
CC> readelf --debug-dump hello
Contents of the .debug_aranges section:

  Length:                   44
  Version:                  2
  Offset into .debug_info:  0x0
  Pointer Size:             8
  Segment Size:             0

    Address            Length
    0000000000400526 000000000000001a
    0000000000000000 0000000000000000

Contents of the .debug_info section:

  Compilation Unit @ offset 0x0:
   Length:        0x8d (32-bit)
   Version:       4
   Abbrev Offset: 0x0
```

Basic background

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Software and Tools

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Goal

- In this lecture we will present the main tools that will be used in the forthcoming modules of Software Security.

Prerequisites

- Lecture:
  - Basic knowledge of C
  - Basic knowledge of Python

Outline

- Reading ELF data
- GDB: The GNU Project Debugger
  - GEF: GDB Enhanced Features
- Pwntools
- Other tools:
  - Radare2
  - Ghidra

Gathering info from binary files

Non entriamo nel dettaglio

Given a binary file we can…

- check if the file is executable or not
- discover the architecture for which the binary has been compiled
- collect symbols and strings used in the program
- check if there is a running process associated with the binary
- read the SHA of a file and check if it is associated with some malicious software
- identify function names and used libraries

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- Pwntools
- Other tools:
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Gathering info from binary files

- Several tools are available to extract information from an ELF file, such as:
  - strings
  - objdump
  - readelf

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  - readelf

Strings

- This is a simple *terminal tool* that permits collecting all the *strings* occurring in a binary file
- For each given file, *strings* print the *printable character sequences* that are *at least 4 characters long* and are followed by an *unprintable character*
- Collected strings can give us info about *secrets* and used data

Strings (an example)

- In a Linux system, we can use *strings* to collect data from */bin/bash*, the *Bash shell:*

```
CC> strings /bin/bash | more
/lib64/ld-linux-x86-64.so.2
 $DJ
CDDB
E %
0`0
"BB1
B8:
0D@kB
 9E4
NR l
 "7$aD
`% H0A
Hap5
@ E<
($B
d> 7
`       0
A!I`
R       2(
        !C)
&51H
```

Strings (some options)

Non entriamo nel dettaglio

- Relevant options are:
  - *–d* (or *--data*), strings are collected only from *data sections*
  - *-n <num>* (or *--bytes=<num>*), prints only strings with *<num>* characters (by default 4)
  - *-h* (or *--help*), prints program help

Gathering info from binary files

- Several tools are available to extract information from an ELF file, such as:
  - strings
  - objdump
  - readelf

objdump

- …displays information about one (or more) object files, such as:
  - information from the overall header of an object file:

```
CC> objdump -f /bin/bash

/bin/bash:     file format elf64-x86-64
architecture: i386:x86-64, flags 0x00000150:
HAS_SYMS, DYNAMIC, D_PAGED
start address 0x0000000000030430
```

objdump

- information from the section headers of the object file (option –*h*):

```
CC> objdump -h /bin/bash

/bin/bash:     file format elf64-x86-64

Sections:
Idx Name          Size      VMA               LMA               File off  Algn
  0 .interp       0000001c  0000000000000318  0000000000000318  00000318  2**0
                  CONTENTS, ALLOC, LOAD, READONLY, DATA
  1 .note.gnu.property 00000020  0000000000000338  0000000000000338  00000338  2**3
                  CONTENTS, ALLOC, LOAD, READONLY, DATA
  2 .note.gnu.build-id 00000024  0000000000000358  0000000000000358  00000358  2**2
                  CONTENTS, ALLOC, LOAD, READONLY, DATA
  3 .note.ABI-tag 00000020  000000000000037c  000000000000037c  0000037c  2**2
                  CONTENTS, ALLOC, LOAD, READONLY, DATA
  4 .gnu.hash     00004aac  00000000000003a0  00000000000003a0  000003a0  2**3
                  CONTENTS, ALLOC, LOAD, READONLY, DATA
  5 .dynsym       0000e418  0000000000004e50  0000000000004e50  00004e50  2**3
                  CONTENTS, ALLOC, LOAD, READONLY, DATA
  6 .dynstr       00009740  0000000000013268  0000000000013268  00013268  2**0
                  CONTENTS, ALLOC, LOAD, READONLY, DATA
  7 .gnu.version  00001302  0000000000001c9a8  0000000000001c9a8  0001c9a8  2**1
```

objdump

- Content of specific sections:

```
CC> objdump -s -j .rodata /bin/bash | more

/bin/bash:     file format elf64-x86-64

Contents of section .rodata:
 de000 01000200 474e5520 62617368 2c207665  ....GNU bash, ve
 de010 7273696f 6e202573 2d282573 290a0078  rsion %s-(%s)..x
 de020 38365f36 342d7063 2d6c696e 75782d67  86_64-pc-linux-g
 de030 6e750047 4e55206c 6f6e6720 6f707469  nu.GNU long opti
 de040 6f6e733a 0a00092d 2d25730a 00536865  ons:...--%s..She
 de050 6c6c206f 7074696f 6e733a0a 00092d25  ll options:...-%
 de060 73206f72 202d6f20 6f707469 6f6e0a00  s or -o option..
 de070 72756e5f 6f6e655f 636f6d6d 616e6400  run_one_command.
 de080 2d630072 62617368 00492068 61766520  -c.rbash.I have
 de090 6e6f206e 616d6521 003f3f68 6f73743f  no name!.??host?
 de0a0 3f004241 53485f45 4e560050 4f534958  ?.BASH_ENV.POSIX
 de0b0 4c595f43 4f525245 43540050 4f534958  LY_CORRECT.POSIX
 de0c0 5f504544 414e5449 43005c73 2d5c765c  _PEDANTIC.\s-\v\
 de0d0 2420003e 20007e2f 2e626173 68726300  $ .> .~/.bashrc.
 de0e0 25733a20 696e7661 6c696420 6f707469  %s: invalid opti
 de0f0 6f6e0025 6325633a 20696e76 616c6964  on.%c%c: invalid
 de100 206f7074 696f6e00 6c6f6769 6e5f7368   option.login_sh
```

objdump

- This tool can be also used to disassemble binaries:

```
CC> objdump -D ./aprogram | grep "main."
    1081:       48 8d 3d c1 00 00 00    lea    0xc1(%rip),%rdi        # 1149 <main>
    1088:       ff 15 52 2f 00 00       callq  *0x2f52(%rip)        # 3fe0 <__libc_start_main@GLIBC_2.2.5>
0000000000001149 <main>:
CC>
```

- A detailed list of functionalities can be obtained by invoking the program with options *–H* (or *--help*)

Gathering info from binary files

- Several tools are available to extract information from an ELF file, such as:
  - strings
  - objdump
  - readelf

readelf

- Main info displayed with *readelf* are:
  - info at the ELF header of the file (option -h)

```
CC> readelf -h /bin/bash
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:                             ELF64
  Data:                              2's complement, little endian
  Version:                           1 (current)
  OS/ABI:                            UNIX - System V
  ABI Version:                       0
  Type:                              DYN (Shared object file)
  Machine:                           Advanced Micro Devices X86-64
  Version:                           0x1
  Entry point address:               0x30430
  Start of program headers:          64 (bytes into file)
  Start of section headers:          1181528 (bytes into file)
  Flags:                             0x0
  Size of this header:               64 (bytes)
  Size of program headers:           56 (bytes)
  Number of program headers:         13
  Size of section headers:           64 (bytes)
  Number of section headers:         30
  Section header string table index: 29
```

readelf

- info available in the segment headers (option –l):

```
CC> readelf -l /bin/bash

Elf file type is DYN (Shared object file)
Entry point 0x30430
There are 13 program headers, starting at offset 64

Program Headers:
  Type           Offset             VirtAddr           PhysAddr
                 FileSiz            MemSiz              Flags  Align
  PHDR           0x0000000000000040 0x0000000000000040 0x0000000000000040
                 0x00000000000002d8 0x00000000000002d8  R      0x8
  INTERP         0x0000000000000318 0x0000000000000318 0x0000000000000318
                 0x000000000000001c 0x000000000000001c  R      0x1
      [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
  LOAD           0x0000000000000000 0x0000000000000000 0x0000000000000000
                 0x000000000002ce70 0x000000000002ce70  R      0x1000
  LOAD           0x000000000002d000 0x000000000002d000 0x000000000002d000
                 0x00000000000b0705 0x00000000000b0705  R E    0x1000
  LOAD           0x00000000000de000 0x00000000000de000 0x00000000000de000
                 0x0000000000036198 0x0000000000036198  R      0x1000
  LOAD           0x0000000000114cf0 0x0000000000115cf0 0x0000000000115cf0
```

readelf

- The entries in symbol table section of the file:

```
CC> readelf -s /bin/bash | more

Symbol table '.dynsym' contains 2433 entries:
   Num:    Value          Size Type    Bind   Vis      Ndx Name
     0: 0000000000000000     0 NOTYPE  LOCAL  DEFAULT  UND
     1: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND endgrent@GLIBC_2.2.5 (2)
     2: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND __ctype_toupper_loc@GLIBC_2.3 (3)
     3: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND iswlower@GLIBC_2.2.5 (2)
     4: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND sigprocmask@GLIBC_2.2.5 (2)
     5: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND __snprintf_chk@GLIBC_2.3.4 (4)
     6: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND free@GLIBC_2.2.5 (2)
     7: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND getservent@GLIBC_2.2.5 (2)
     8: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND wcscmp@GLIBC_2.2.5 (2)
     9: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND putchar@GLIBC_2.2.5 (2)
    10: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND tputs@NCURSES6_TINFO_5.0.19991023 (5)
    11: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND strcasecmp@GLIBC_2.2.5 (2)
    12: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND localtime@GLIBC_2.2.5 (2)
    13: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND mblen@GLIBC_2.2.5 (2)
    14: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND __vfprintf_chk@GLIBC_2.3.4 (4)
    15: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND abort@GLIBC_2.2.5 (2)
    16: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND __errno_location@GLIBC_2.2.5 (2)
    17: 0000000000000000     0 FUNC    GLOBAL DEFAULT  UND strncpy@GLIBC_2.2.5 (2)
```

Outline

- Reading ELF data
- GDB: The GNU Project Debugger
  - GEF: GDB Enhanced Features
- Pwntools
- Other tools:
  - Radare2
  - Ghidra

GDB: The GNU Project Debugger

- GDB is a debugging tool that can be used to…
  - Start your program, specifying anything that might affect its behavior
  - Stop your program at the occurrence of specified conditions
  - Examine what happened, when your program stopped
  - Change memory or registers content while your program is running

GDB: The GNU Project Debugger

- GDB supports multiple languages:
  - Assembly (x86, ARM, MIPS…)
  - C
  - C++
  - Rust
  - …

GDB: The GNU Project Debugger

- GDB is a terminal tool and can be launched by executing program *gdb*

```
CC> gdb
GNU gdb (Ubuntu 9.2-0ubuntu1~20.04) 9.2
Copyright (C) 2020 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word".
(gdb)
```

GDB: The GNU Project Debugger

- You can invoke *gdb* by passing the program to debug

```
gdb <program>
```

- you can pass the *process ID* as a second argument to debug a running process:

```
gdb <program> <pid>
```

- Or equivalently use:

```
gdb -p <pid>
```

GDB: Startup

Non entriamo nel dettaglio

When executed, gdb performs the following main steps:

- sets up the command interpreter as specified by the command line
  - *gdb* can be executed in different *modes:*
    - batch: executes a list of commands and waits for termination
    - quiet: does not print introductory and copyright messages
- Load configuration files
  - Defines the behaviour of gdb at startup
  - Can be global at the system level or local to the current directory
  - Details are available at the gdb documentation page
- Load symbols of debugged program
- Waits for user commands

GDB: Commands

- A *gdb* command consists of a single line of input, containing:
  - a *command name*
  - a sequence of *parameters*

- Command *run* (or *r*) can be used to *start* a program in *gdb:*

```
CC> gdb -q aprogram
Reading symbols from aprogram...
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/aprogram
Hello World!
[Inferior 1 (process 36198) exited normally]
(gdb)
```

GDB: Commands

- Command *help* can be used to access the list of available commands:

```
(gdb) help
List of classes of commands:

aliases -- Aliases of other commands.
breakpoints -- Making program stop at certain points.
data -- Examining data.
files -- Specifying and examining files.
internals -- Maintenance commands.
obscure -- Obscure features.
running -- Running the program.
stack -- Examining the stack.
status -- Status inquiries.
support -- Support facilities.
tracepoints -- Tracing of program execution without stopping the program.
user-defined -- User-defined commands.

Type "help" followed by a class name for a list of commands in that class.
Type "help all" for the list of all commands.
Type "help" followed by command name for full documentation.
Type "apropos word" to search for commands related to "word".
Type "apropos -v word" for full documentation of commands related to "word".
Command name abbreviations are allowed if unambiguous.
```

GDB: Commands

- Commands *set args* and *show args* can be used to set and show program arguments:

```
CC> gdb -q aprogram
Reading symbols from aprogram...
(gdb) set args 10 test /usr/bin
(gdb) show args
Argument list to give program being debugged when it is started is "10 test /usr/bin".
(gdb)
```

GDB: Commands

Non entriamo nel dettaglio

- The *environment* consists of a set of *environment variables* storing info such as *user name*, *home directory*, *terminal type*, or the search *path* for programs to run.
- Environment variables can be accessed/changed within *gdb* via the following commands:
  - *path <directory>* add the directory to the path
  - *show paths* show the content of *path*
  - *show environment [<var>]* show the environment (or the specific variable)
  - *set environment <var> <value>* set value for the given variable
  - *unset environment <var>* delete the given variable from the environment

GDB: Stopping and Continuing

- The principal reason to use a debugger is that we can stop a program before it terminates to check its status and, if we experience some problems, investigate and find out why.
- Inside *gdb,* a program may stop for:
  - a *breakpoint*
  - a *signal*
  - the completion of the execution of a *step*

GDB: Breakpoints

- A *breakpoint* makes your program stop whenever a certain point in the program is reached
  - details can be added to the breakpoint to control in finer detail whether your program stops
- A *watchpoint* is a special breakpoint that stops your program when the value of an expression changes
- A *catchpoint* is another special breakpoint that stops your program when a certain kind of event occurs (for example, a signal or an error)

GDB: Breakpoints

- Breakpoints are set with the *break* command (abbreviated *b*):
  - *break location* set break point at the given location
  - *break* set break point at the next instruction
  - *break [location] if <cond>* set break point with the given condition

- A breakpoint location can be:
  - A line number
  - A function name
  - An address

GDB: Example

- Let us consider the following simple c program:

```
#include <stdio.h>
#include <stdlib.h>


int fibonacci(int n);

int main(int argc , char** argv) {
     int n = 10;
     if (argc>1) {
        n = atoi(argv[1]);
     }
     printf("fib(%d) is %d\n",n,fibonacci(n));
}

int fibonacci(int n) {
    int k = n;
    if (k<=2) {
        return 1;
    } else {
        return fibonacci(n-1)+fibonacci(n-2);
    }
}
fibonacci.c (END)
```

GDB: Example

Example (see next slides)

```
CC> gcc -g -o fibonacci fibonacci.c
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) set args 5
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci 5

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

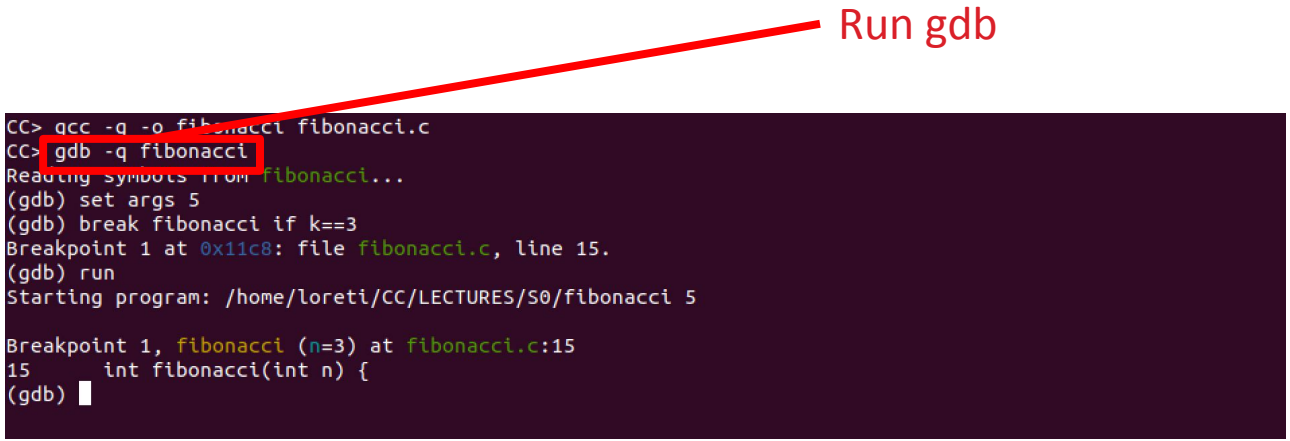
GDB: Example

We compile it for debugging

```
CC> gcc -g -o fibonacci fibonacci.c
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) set args 5
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci 5

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

GDB: Example



GDB: Example

Set program arguments

```
CC> gcc -g -o fibonacci fibonacci.c
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) set args 5
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci 5

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

GDB: Example


```
CC> gcc -g -o fibonacci fibonacci.c
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) set args 5
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11ca: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci 5

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Add a conditional breakpoint

GDB: Example


```
CC> gcc -g -o fibonacci fibonacci.c
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) set args 5
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci 5

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Run our program

GDB: Example


```
CC> gcc -g -o fibonacci fibonacci.c
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) set args 5
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci 5

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Computation is stopped when the breakpoint with the specific condition is reached

GDB: Continue and Stepping

- When a program stops at a breakpoint, its execution can be resumed exploiting 2 functionalities:
  - *Continuing* means resuming program execution until your program completes normally
  - *Stepping* means executing just one more “step” of your program
    - A step can be either an instruction of source code, or an instructions of the assembly

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break main
Breakpoint 1 at 0x1169: file fibonacci.c, line 7.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, main (argc=21845, argv=0x0) at fibonacci.c:7
7       int main(int argc , char** argv) {
(gdb) step
8               int n = 10;
(gdb) step
9               if (argc>1) {
(gdb) continue
Continuing.
fib(10) is 55
[Inferior 1 (process 36526) exited normally]
(gdb)
```

Example (see next slides)

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break main
Breakpoint 1 at 0x1169: file fibonacci.c, line 7.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, main (argc=21845, argv=0x0) at fibonacci.c:7
7       int main(int argc , char** argv) {
(gdb) step
8           int n = 10;
(gdb) step
9           if (argc>1) {
(gdb) continue
Continuing.
fib(10) is 55
[Inferior 1 (process 36526) exited normally]
(gdb)
```

A single *step* is executed, *gdb* prints the *next* instruction

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break main
Breakpoint 1 at 0x1169: file fibonacci.c, line 7.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, main (argc=21845, argv=0x0) at fibonacci.c:7
7       int main(int argc , char** argv) {
(gdb) step
8           int n = 10;
(gdb) step
9           if (argc>1) {
(gdb) continue
Continuing.
fib(10) is 55
[Inferior 1 (process 36526) exited normally]
(gdb)
```

We can step again…

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break main
Breakpoint 1 at 0x1169: file fibonacci.c, line 7.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, main (argc=21845, argv=0x0) at fibonacci.c:7
7       int main(int argc , char** argv) {
(gdb) step
8           int n = 10;
(gdb) step
9           if (argc>1) {
(gdb) continue
Continuing.
fib(10) is 55
[Inferior 1 (process 36526) exited normally]
(gdb)
```

…or continue until the program terminates or another breakpoint is reached.

GDB: Inspecting the stack

- When your program has stopped, the first thing we need to know is where it *stopped* and *how* it got there
- The first thing to consider is the content of the *stack*
- gdb commands are available to examine the stack and to read the content of the *stack* and to read any of the stored *stack frames*
- In the *stack* frame you can find:
  - the location of the call in your program
  - the arguments of the call
  - the local variables of the function being called

GDB: Inspecting the stack

- The following commands can be used to read the content of the stack:
  - *frame [<selection>]*
    - prints a brief description of the selected stack frame.
  - *info frame [<selection>]*
    - prints a verbose description of the selected stack frame

- See *gdb* documentation for a (long) list of options

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb) frame
#0  fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Example (see next slides)

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb) frame
#0  fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Command *frame* is used to obtain a brief description of the current frame

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb) frame
#0  fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Number of current frame

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb) frame
#0  fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Function name, its arguments and code line

GDB: Example


```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break fibonacci if k==3
Breakpoint 1 at 0x11c8: file fibonacci.c, line 15.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb) frame
#0  fibonacci (n=3) at fibonacci.c:15
15      int fibonacci(int n) {
(gdb)
```

Source code

GDB: Example

- Command *info frame* permits getting detailed info about the current stack frame:

```
(gdb) info frame
Stack level 0, frame at 0x7fffffffdcd0:
 rip = 0x5555555551c8 in fibonacci (fibonacci.c:15); saved rip = 0x555555555207
 called by frame at 0x7fffffffdd10
 source language c.
 Arglist at 0x7fffffffdcc0, args: n=3
 Locals at 0x7fffffffdcc0, Previous frame's sp is 0x7fffffffdcd0
 Saved registers:
  rip at 0x7fffffffdcc8
(gdb)
```

GDB: Example

Non entriamo nel dettaglio

(gdb) info frame
Stack level 0, frame at 0xb75f7390:
eip = 0x804877f in base::func() (testing.cpp:16); saved **eip** 0x804869a
called by frame at 0xb75f73b0
source language c++.
Arglist at 0xb75f7388, args: this=0x0
Locals at 0xb75f7388, Previous frame's sp is 0xb75f7390
Saved registers:
ebp at 0xb75f7388, eip at 0xb75f738c

- the address of the frame
- the address of the next frame down (called by this frame)
- the address of the next frame up (caller of this frame)
- the language in which the source code corresponding to this frame is written
- the address of the frame's arguments
- the address of the frame's local variables
- the program counter saved in it (the address of execution in the caller frame)
- which registers were saved in the frame

https://stackoverflow.com/questions/5144727/how-to-interpret-gdb-info-frame-output

saved eip "0x804869a" is the so called "return address", i.e., the instruction to resume in caller stack frame after returning from this callee stack.

GDB: Disassembly

- Command *disas* can be used to *disassemble* a given function

```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break main
Breakpoint 1 at 0x1169: file fibonacci.c, line 7.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, main (argc=21845, argv=0x0) at fibonacci.c:7
7       int main(int argc , char** argv) {
(gdb) disas
Dump of assembler code for function main:
=> 0x0000555555555169 <+0>:     endbr64
   0x000055555555516d <+4>:     push   %rbp
   0x000055555555516e <+5>:     mov    %rsp,%rbp
   0x0000555555555171 <+8>:     sub    $0x20,%rsp
   0x0000555555555175 <+12>:    mov    %edi,-0x14(%rbp)
   0x0000555555555178 <+15>:    mov    %rsi,-0x20(%rbp)
   0x000055555555517c <+19>:    movl   $0xa,-0x4(%rbp)
   0x0000555555555183 <+26>:    cmpl   $0x1,-0x14(%rbp)
   0x0000555555555187 <+30>:    jle    0x55555555519f <main+54>
   0x0000555555555189 <+32>:    mov    -0x20(%rbp),%rax
   0x000055555555518d <+36>:    add    $0x8,%rax
   0x0000555555555191 <+40>:    mov    (%rax),%rax
   0x0000555555555194 <+43>:    mov    %rax,%rdi
   0x0000555555555197 <+46>:    callq  0x555555555070 <atoi@plt>
   0x000055555555519c <+51>:    mov    %eax,-0x4(%rbp)
   0x000055555555519f <+54>:    mov    -0x4(%rbp),%eax
   0x00005555555551a2 <+57>:    mov    %eax,%edi
   0x00005555555551a4 <+59>:    callq  0x5555555551c8 <fibonacci>
   0x00005555555551a9 <+64>:    mov    %eax,%edx
   0x00005555555551ab <+66>:    mov    -0x4(%rbp),%eax
   0x00005555555551ae <+69>:    mov    %eax,%esi
   0x00005555555551b0 <+71>:    lea    0xe4d(%rip),%rdi        # 0x555555556004
--Type <RET> for more, q to quit, c to continue without paging--
```

GDB: Examining data

- The usual way to examine data in your program is with the *print* command (abbreviated *p*), or its synonym *inspect*

```
CC> gdb -q fibonacci
Reading symbols from fibonacci...
(gdb) break main
Breakpoint 1 at 0x1169: file fibonacci.c, line 7.
(gdb) run
Starting program: /home/loreti/CC/LECTURES/S0/fibonacci

Breakpoint 1, main (argc=21845, argv=0x0) at fibonacci.c:7
7       int main(int argc , char** argv) {
(gdb) print n
$1 = 0
(gdb) step
8               int n = 10;
(gdb) print n
$2 = 0
(gdb) step
9               if (argc>1) {
(gdb) print n
$3 = 10
(gdb)
```

Outline

- Reading ELF data
- GDB: The GNU Project Debugger
  - GEF: GDB Enhanced Features
- Pwntools
- Other tools:
  - Radare2
  - Ghidra

GEF: GDB Enhanced Features

Non entriamo nel dettaglio

- GEF consists of a set of commands that extends GDB with additional features for *dynamic analysis* and *exploit development*.
- GEF is based on GDB Python API
- Main GEF features:
  - Embedded hexdump view
  - Automatic dereferencing of *data* and *registers*
  - Heap analysis
  - Display ELF information
- Detailed GEF documentation is available at

  https://gef.readthedocs.io/en/master/

GEF: GDB Enhanced Features

Non entriamo nel dettaglio

- GEF has been designed with the following constraints:
  - Simple and fast to install
  - No external dependency
    - Some extensions are available to increase available features
  - Compatible with both Python2 and Python3
  - Abstract from specific architecture
  - Extensible
  - Well documented

GEF: GDB Enhanced Features

- Pretty printing of registers (with automatic dereferencing)…

```
gef➤  registers
$zero: 0x0
$at   : 0x1
$v0   : 0x77ff9490
$v1   : 0x7fffe6c8  →  0x77e61498  →  <__libc_start_main+200> bnez v0, 0x77e614f0 <__libc_start_main+288>
$a0   : 0x1
$a1   : 0x7fffe784  →  0x7fffe880  →  "/home/user/simple-bof"
$a2   : 0x7fffe78c  →  0x7fffe896  →  "LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so[...]"
$a3   : 0x0
$t0   : 0x77e613d0  →  <__libc_start_main+0> lui gp, 0x17
$t1   : 0x80808080
$t2   : 0xb64
$t3   : 0x0
$t4   : 0x3
$t5   : 0x0
$t6   : 0x77fff7f0  →  0x77fcc000  →  0x464c457f
$t7   : 0x55555704  →  <hlt+0> b 0x55555704 <hlt>
$s0   : 0x0
$s1   : 0x555559f0  →  <__libc_csu_init+0> lui gp, 0x2
$s2   : 0x77ffeedc  →  0xbafeb100
$s3   : 0x0
```

GEF: GDB Enhanced Features

- Info about the *heap chunks…*

```
gef➤  heap chunks
Chunk(addr=0x555555756010, size=0x250, flags=PREV_INUSE)
    [0x0000555555756010     00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00    ................]
Chunk(addr=0x555555756260, size=0x110, flags=PREV_INUSE)
    [0x0000555555756260     61 61 61 61 61 61 61 61 61 61 61 61 61 61 61 61    aaaaaaaaaaaaaaaa]
Chunk(addr=0x555555756370, size=0x110, flags=PREV_INUSE)
    [0x0000555555756370     62 62 62 62 62 62 62 62 62 62 62 62 62 62 62 62    bbbbbbbbbbbbbbbb]
Chunk(addr=0x555555756480, size=0x110, flags=PREV_INUSE)
    [0x0000555555756480     63 63 63 63 63 63 63 63 63 63 63 63 63 63 63 63    cccccccccccccccc]
Chunk(addr=0x555555756590, size=0x110, flags=PREV_INUSE)
    [0x0000555555756590     64 64 64 64 64 64 64 64 64 64 64 64 64 64 64 64    dddddddddddddddd]
Chunk(addr=0x5555557566a0, size=0x20970, flags=PREV_INUSE)  ←  top chunk
```

GEF: GDB Enhanced Features

- Security info…

```
gef➤  checksec
[+] checksec for '/home/user/mips-stack-bof'
Canary                            : No
NX                                : No
PIE                               : Yes
Fortify                           : No
RelRO                             : No
gef➤
```

GEF: GDB Enhanced Features

- Hexdump view of a memory range…

```
gef➤  db $sp
0x00007fffffffdfa0     60 62 75 55 55 55 00 00 70 63 75 55 55 55 00 00     `buUUU..pcuUUU..
0x00007fffffffdfb0     80 64 75 55 55 55 00 00 90 65 75 55 55 55 00 00     .duUUU...euUUU..
0x00007fffffffdfc0     30 47 55 55 55 55 00 00 97 5b a0 f7 ff 7f 00 00     0GUUUU...[......
0x00007fffffffdfd0     01 00 00 00 00 00 00 00 a8 e0 ff ff ff 7f 00 00     ................
```

GEF: GDB Enhanced Features

Non entriamo nel dettaglio

- GDB extensions:
  - *keystone-assemble* Assembly engine
  - *capstone-disassemble* Disassembly engine
  - *unirorn-emulate* Emulation engine
  - *Ropper* ROP Gadget generator
  - *RetDec* Decompiler

(see GEF docs for details)

Outline

- Reading ELF data
- GDB: The GNU Project Debugger
  - GEF: GDB Enhanced Features
- Pwntools (visto brevemente nel laboratorio introduttivo Python)
- Other tools:
  - Radare2
  - Ghidra

Andremo veloci sui tool aggiuntivi. Se vi dovesse servire, fate riferimento alle slide e/o alla documentazione degli strumenti.

Pwntools…

- …is a *CTF framework* and *exploit development library* that is…
  - written in Python
  - designed for rapid prototyping
  - intended to make exploit writing as simple as possible

Pwntools: Tubes

- *Tubes* are I/O wrappers supporting the main type of connections:
  - Local processes (pipe)
  - Remote TCP or UDP connections
  - Process running on a remote server over SSH
  - Serial port I/O

Pwntools: Tubes

- Via a *tube* one can:
  - Send data
    - *send(data)* Sends data
    - *sendline(line)* Sends data plus a newline
  - Receive data
    - *recv(n)* Receive the given number of bytes
    - *recvline()* Receive data until a newline is found
    - *recvuntil(delim)* Receive data until a delimiter is found
    - *recvregex(pattern)* Receive data until a regex pattern is satisfied
    - *recvrepeat(timeout)* Keep receiving data until a timeout occurs
    - *clean()* Discard all vuffered data
  - Manipulating integers
    - *pack(int)* Sends a word-size packed integer
    - *unpack()* Receives and unpacks word-size integer

Pwntools: Processes

- A *tube* can be used to interact with a *process* that can be created by passing:
  - the name of the binary to execute

```
io = process('sh')
```

  - the list of arguments and environment

```
io = process(['sh', '-c', 'echo $MYENV'], env={'MYENV': 'MYVAL'})
```

Pwntools: Example

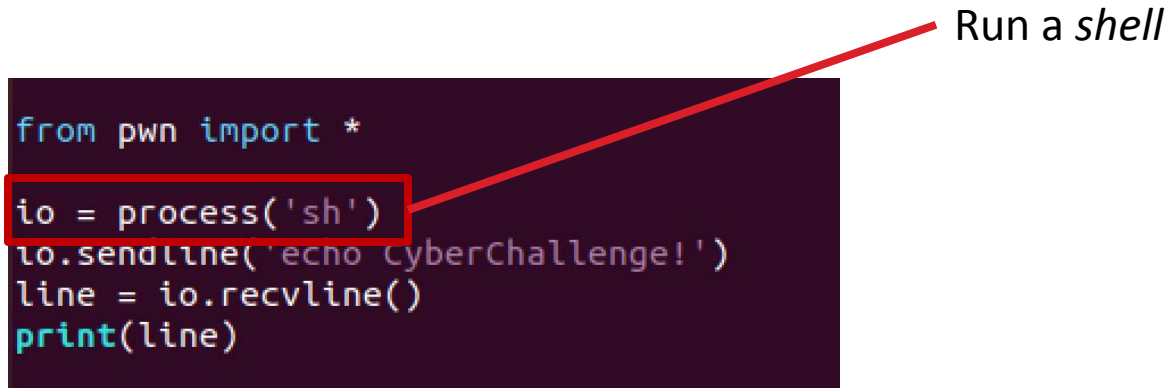
- Let us consider the following *python* script:

```
from pwn import *

io = process('sh')
io.sendline('echo CyberChallenge!')
line = io.recvline()
print(line)
```

Pwntools: Example

- Let us consider the following *python* script:



Pwntools: Example

- Let us consider the following *python* script:

```
from pwn import *

io = process('sh')
io.sendline('echo CyberChallenge!')
line = io.recvline()
print(line)
```

Send a command

Pwntools: Example

- Let us consider the following *python* script:

```
from pwn import *

io = process('sh')
io.sendline('echo CyberChallenge!')
line = io.recvline()
print(line)
```

Receive the result

Pwntools: Networking

- Via pwntools one can easily create network connections or waiting for incoming ones:
  - *remote* is used to open a connection with a remote server
  - *listen* is used to wait for an incoming connection

- Both *tcp* and *udp* protocols can be used

Pwntools: Example

- Let us consider the following *python* script:

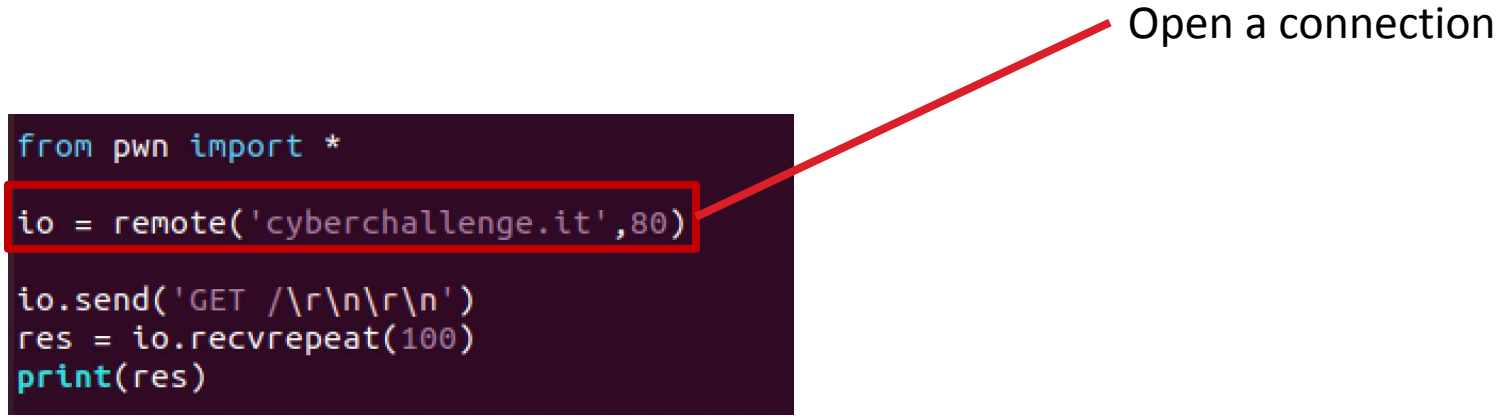
```
from pwn import *

io = remote('cyberchallenge.it',80)

io.send('GET /\r\n\r\n')
res = io.recvrepeat(100)
print(res)
```

Pwntools: Example

- Let us consider the following *python* script:



Pwntools: Example

- Let us consider the following *python* script:

```
from pwn import *

io = remote('cyberchallenge.it',80)

io.send('GET /\r\n\r\n')
res = io.recvrepeat(100)
print(res)
```

Send a request

Pwntools: Example

- Let us consider the following *python* script:

```
from pwn import *

io = remote('cyberchallenge.it',80)

io.send('GET /\r\n\r\n')
res = io.recvrepeat(100)
print(res)
```

Receive a response (with timeout 100 seconds)

Pwntools: SSH

- *Pwntools* supports SSH sessions:

```
session = ssh('username', 'hostname', password='password')
```

- Given a session, one can execute *processes* as we have already seen in the previous slides:

```
io = session.process('sh')
```

Pwntools: Other Features

- A *utility* module with functions for
  - *Packaging* and *unpackaging* integers
  - File I/O
  - Hashing and Encoding
  - Pattern Generation
- ELF files manipulation
- Assembling and disassembling

Outline

- Reading ELF data
- GDB: The GNU Project Debugger
  - GEF: GDB Enhanced Features
- Pwntools
- Other tools:
  - Radare2
  - Ghidra

Radare2

- Radare2 is a *toolchain* for easing task like:
  - Software Reverse Engineering
  - Exploiting
  - Debugging
- Radare2 is *open source* available at

  https://github.com/radareorg/radare2

Radare2

- The framework consists of a number of *command-line* tools, among which we mention:
  - *radare2:* a powerful *hexadecimal editor* and *debugger*
  - *rabin2:* a program that permits extracting information from executable binaries
  - *rasm2:* a command line *assembler* and *disassembler*
  - *rahash2:* an implementation of a *block-based* hash tool

Outline

- Reading ELF data
- GDB: The GNU Project Debugger
  - GEF: GDB Enhanced Features
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  - Ghidra

Ghidra

- Ghidra is a Software Reverse Engineering (SRE) tool developed by *National Security Agency* (NSA)
- Ghidra is a *platform independent* and includes:
  - an interactive *disassembler* and *decompiler*
  - a set of tools to support *code analysis*
- The tool is *open source* and available at:
  - https://ghidra-sre.org
- More details about Ghidra and its use will be given in the next module (Software Security 1)

Software and Tools

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https://cybersecnatlab.it

Identifying vulnerabilities

- Information collected from binaries can be used to discover program vulnerabilities:
  - where a program takes an input and if this is validated
  - if memory is safely managed
  - if up-to-date libraries or third-party frameworks are used
  - how the application is compiled
  - if (static) strings are used to record sensitive data

Disassembly in action


```c
#include <stdio.h>

#define MESSAGE "Hello world!"

int main() {
    printf(MESSAGE);
    return 0;
}
```

gcc –s hello.c

```
main:
.LFB0:
        .cfi_startproc
        pushq   %rbp
        .cfi_def_cfa_offset 16
        .cfi_offset 6, -16
        movq    %rsp, %rbp
        .cfi_def_cfa_register 6
        movl    $.LC0, %edi
        movl    $0, %eax
        call    printf
        movl    $0, %eax
        popq    %rbp
        .cfi_def_cfa 7, 8
        ret
```

gcc –o hello hello.c

hello.o

objdump -d hello

```
0000000000400526 <main>:
  400526:       55                      push   %rbp
  400527:       48 89 e5                mov    %rsp,%rbp
  40052a:       bf c4 05 40 00          mov    $0x4005c4,%edi
  40052f:       b8 00 00 00 00          mov    $0x0,%eax
  400534:       e8 c7 fe ff ff          callq  400400 <printf@plt>
  400539:       b8 00 00 00 00          mov    $0x0,%eax
  40053e:       5d                      pop    %rbp
  40053f:       c3                      retq
```

Basic Disassembly Algorithm

- The following steps are executed by a disassembler:
  - Identify a region of code to disassemble
  - Given the address of an instruction match its binary opcode value to the assembly language mnemonics (MOV, ADD, CMP, JMP, …)
  - Format the reconstructed code
  - Move to the next instruction in the file

Disassembly at work: An Example

- Let us consider the program *findmysecret*:

```
CC> ./findmysecret
You will never know my secret!
CC>
```

- We know that a secret is stored somewhere in the program and we must discover it

Disassembly at work: An Example

- We can use *gdb* to call one of the program's functions
- First, we must load our program and set a break point in the main:

```
gef➤  file findmysecret
Reading symbols from findmysecret...
(No debugging symbols found in findmysecret)
gef➤  break main
Breakpoint 1 at 0x1234
gef➤  run
```

- After that we can call the function and see the secret:

```
trace
[#0] 0x56556234 →main()

gef➤  call (void *) functionOne()
My secret is 42!

$2 = (void *) 0x12
gef➤
```

GDB Cheat sheet

- break *<indirizzo> or <nome>
- run <args>
- info functions → mostra le funzioni all'interno del codice e il loro indirizzo in memoria
- info registers → mostra lo stato dei registri in quel momento
- set $<registro> = <valore> → setta il valore del registro
- x → stampare indirizzi di memoria
  - x <indirizzo o $registro> → stampa il valore in quell'indirizzo o registro
  - x/<formato> <indirizzo o $registro> → stampa il valore in quell'indirizzo o registro come:

    s = stringa / **x** = esadecimale / **d** = decimale / **i** = istruzione ASM

    Esempio: x/s $rax

GDB Cheat sheet

- x/<num_elem><formato><modificatore> <indirizzo o $registro> → permette di specificare più opzioni oltre al formato:
  - num elem → numero di elementi da stampare
  - modificatore → b = byte / h = halfword (16 bit) / w = word (32-bit) / g = giant word (64-bit)

    Esempio: x/3xw $esp
- x/<numero>i $pc → mostrare in ASM qual è l'istruzione corrente e quali saranno le prossime <numero> - 1
- $pc = program counter, alias di $rip
- p (alias di print e di inspect)
  - print → mostra il valore di variabili, indirizzi di memoria e registri
- stepi → "step into" una funzione, entrare dentro una funzione

https://darkdust.net/files/GDB%20Cheat%20Sheet.pdf
https://cs.brown.edu/courses/cs033/docs/guides/gdb.pdf